International Institute of Information Technology Hyderabad

The Project Report

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Title: *Automated Weapon Detection in Banks and ATMs Using YOLOv8*

Abstract

In this report, we present a custom object detection system for automated weapon detection in security-sensitive environments, specifically in banks and ATMs. Utilizing the YOLOv8 object detection model, the system is designed to identify potential threats, including knives, handguns, and long guns, in real-time video feeds. The proposed method involves creating a custom dataset, annotating it using the Label- studio tool, and training the YOLOv8 model on Google Colab. The trained model is then implemented in Visual Studio Code (VS Code) for real-time object detection. This study demonstrates the potential of deep learning models for enhancing security and preventing crime in high-risk areas. Results show promising detection accuracy, making it suitable for real-time deployment in surveillance systems.

Introduction

The increasing number of violent crimes, especially armed robberies, necessitates enhanced security at banks and ATMs to protect both the integrity of financial institutions and the safety of their clients and staff. Traditional security measures rely on human operators who monitor video feeds, but this approach can lead to delayed responses due to human fatigue and error.

Deep learning and computer vision offer promising advancements in automating security. This study leverages YOLOv8, a cutting-edge model in the YOLO family, for real-time weapon detection. YOLOv8 is known for its accuracy and speed, making it ideal for security applications. By training the model on a custom dataset of weapon images, the system aims to detect threats—such as guns or knives—in surveillance footage from banks and ATMs. The model is deployed in Visual Studio Code (VS Code) to provide instant alerts and real-time analysis, reducing dependency on human operators and enhancing overall security. The study results demonstrate that YOLOv8 effectively improves detection accuracy and response times, creating a more secure environment in financial institutions.

Objective

1. **Develop a Custom Weapon Detection Dataset**: Create and annotate a dataset of weapon images, including various types of firearms and bladed weapons, to train the model for accurate detection.
2. **Train and Deploy YOLOv8 for Real-Time Detection**: Implement the YOLOv8 model to detect and classify weapons within live surveillance video feeds with high precision and speed.
3. **Evaluate System Performance and Security Effectiveness**: Assess the model's accuracy, precision, recall, and false alarm rate to ensure reliable detection capabilities in high-risk environments.
4. **Integrate Automated Alerting and Real-Time Monitoring**: Enable instant alerts upon weapon detection to aid rapid response and improve security operations by reducing reliance on human monitoring.

Literature Review

Overview of Object Detection Techniques

1. **Early Techniques (SVM & HOG):**
    - Utilized hand-crafted features like edges and gradients for object detection.
    - Employed Support Vector Machines (SVM) for classification.
    - High computational cost and limited effectiveness for real-time applications due to sliding window techniques.
2. **Rise of CNNs:**
    - Convolutional Neural Networks (CNNs) enabled automatic feature extraction, removing the need for manual engineering.
    - Capable of learning complex, hierarchical features directly from image data.
3. **Faster R-CNN (Ren et al., 2015):**
    - Introduced Region Proposal Network (RPN) for generating object locations within the network.
    - Combined object classification and localization for improved detection accuracy.
    - Effective in accuracy but not fully optimized for real-time performance.
4. **SSD (Single Shot MultiBox Detector, Liu et al., 2016):**
    - Eliminated the need for separate region proposals by predicting bounding boxes and classes in a single pass.
    - Used multi-scale feature maps, enhancing detection of objects at various scales.
    - Achieved a balance between speed and precision, making it suitable for real-time tasks.
5. **YOLO (You Only Look Once, Redmon et al., 2016):**
    - Treated object detection as a single regression problem, processing the full image at once.
    - Enabled real-time detection by predicting bounding boxes and class probabilities directly.
    - Successive versions, like YOLOv3 with multi-scale detection, improved accuracy and robustness.
6. **Advancements in YOLO (YOLOv5 & YOLOv8):**
    - YOLOv5 became popular for its ease of use and real-world performance despite not being officially from the original creators.
    - YOLOv8 introduced further architectural and training improvements, achieving even greater speed and precision.
    - Continues to be widely adopted for real-time applications in various fields due to its efficiency and accuracy balance.

Methodology

This section outlines the methodology used to develop the weapon detection system, including data collection and annotation, model training, and implementation.

- **Data Collection**
  The dataset used in this project comprises images containing weapons such as knives, handguns, and long guns. The images were sourced from publicly available datasets as well as custom-collected images to create a more diverse and robust dataset. The dataset is divided into training and validation sets, with a recommended 80-20 split for effective training and evaluation.

  The task of dataset construction and collection was very important and tough as well because there was no benchmark dataset available for this. Dataset for real-time detection was collected and constructed in different phases and data was collected from the internet, extracted from YouTube CCTV videos, through GitHubrepositories, data by the University of Granada research group, and internet movies.

- **WEAPON DATASET CLASSES**

  The weapon dataset for real-time weapon detection is divided into the following several classes:

  1. Pistol
  2. Machine Guns
  3. Knives
  4. Non-weaponed Element

- **Data Annotation**
  To annotate the dataset, we used Label-studio, an open-source graphical annotation tool. The images were annotated by drawing bounding boxes around the objects of interest (weapons), with labels corresponding to the type of weapon (e.g., knife, handgun, long gun). Annotations were saved in YOLO format (TXT), which is compatible with the YOLOv8 model. Proper annotation is crucial for the model's performance, and care was taken to ensure accuracy in the bounding box placements.

Image Class	Input Image	Annotated Image (Label-Studio)
Machine Gun		
Non-Weaponed Element		
Pistol		
Knives		

- **Training the YOLOv8 Model**

  The YOLOv8 model was trained on Google Colab, a cloud-based platform that offers access to powerful GPUs, making it perfect for deep learning applications. The training procedure was painstaking, beginning with the environment setup, which involved configuring Google Colab and installing the relevant libraries, including the ultralytics library, which is required for implementing the YOLOv8 model.

  The next step in dataset preparation was to upload the annotated photos and labels to Google Drive. These were then integrated to the Colab environment, allowing for easy access during training. The model configuration stage was crucial, as the YOLOv8 configuration file, yolov8n.yaml, was customised to match the dataset. This customisation involves selecting the number of classes and altering various hyperparameters to improve the model's performance on the given dataset.

  The training process took 30 epochs, and the model's performance was continuously tracked using important measures like loss, accuracy, and mean Average Precision (mAP). During training, several data augmentation techniques, such as image flipping and scaling, were used to improve the model's capacity to generalise across varied contexts. These augmentations assisted the model in learning more robust features, resulting in improved performance on previously unseen data.

  After finishing the training, the best-performing model weights were saved as best.pt. This file contains the trained model's parameters and is suitable for deployment in real-world applications where the YOLOv8 model can perform tasks such as object detection with high accuracy. The entire procedure demonstrates the power and flexibility of utilising Google Colab to teach advanced machines.

- **Implementation in Visual Studio Code**

  Once the model was trained, it was deployed in a real-time application using Visual Studio Code. The implementation involved the following steps:

1. **Setting Up VS Code Environment:** A Python virtual environment was created, and the necessary libraries were installed.
2. **Loading the Trained Model:** The trained YOLOv8 model (best.pt) was loaded into a Python script within VS Code.
3. **Object Detection:** The script was designed to take input images or video streams and perform object detection using the loaded model. Detection results were displayed with bounding boxes around the detected weapons.

- **Detection Examples**

  The following images showcase examples of weapon detection performed by the trained YOLOv8 model results emphasize the YOLOv8 model's strong performance, ability to accurately detect weapons in various scenarios, highlighting its robustness for real-world applications and reliability in various challenging real-world situations, making it a suitable choice for integration into live surveillance systems for enhanced security monitoring.

  The YOLOv8 model exhibited impressive real-time detection capabilities, processing video streams with minimal latency. This performance makes the model suitable for deployment in live surveillance systems, where rapid response to detected threats is crucial.

Results

The YOLOv8 model demonstrated strong performance in detecting weapons in both static images and video streams. The model achieved an mAP (mean average precision) of over 80% on the validation set, indicating high accuracy in identifying knives, handguns, and long guns. The real-time detection capability of YOLOv8 ensured that the model could be deployed in live surveillance systems without significant latency..

Conclusion

The automated weapon detection system using the YOLOv8 model highlights the potential of AI-driven surveillance to significantly enhance security measures in sensitive environments like banks and ATMs. The high accuracy and real-time detection capabilities demonstrated by the model suggest that it could play a crucial role in proactively identifying and mitigating potential threats, leading to faster response times and a reduction in the need for continuous human monitoring.

The study also underscores the importance of developing and training models on custom datasets tailored to specific security needs, as this approach maximizes the model's effectiveness in real-world scenarios. Additionally, the use of accessible platforms like Google Colab for training illustrates the practicality of deploying such advanced systems without requiring extensive computational resources.

Future directions could include integrating this automated detection system with existing security infrastructure, such as alarm systems and law enforcement communication networks, to create a comprehensive and cohesive security ecosystem. Moreover, expanding the model's capabilities to detect a wider range of suspicious objects or behaviors could further enhance its utility, ensuring that it remains adaptable to evolving security challenges.

Overall, this research provides a solid foundation for the deployment of automated weapon detection systems in high-risk environments, promising significant improvements in safety and security. Moreover, the development of user-friendly interfaces for security personnel to interact with these systems will be crucial in maximizing the practical utility of the technology. As deep learning continues to advance, the potential for automated weapon detection systems to become a standard component of security protocols in banks, ATMs, and other sensitive locations is becoming increasingly feasible.

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Colab Link:

https://colab.research.google.com/drive/1_uEzCEIaMPsJrsNrwNt5CUBaAXV0klhL?usp=sharing